

AWS Key Pair & Subnets — Brief Notes

1. AWS Key Pair

An **AWS Key Pair** is a pair of cryptographic keys used to securely authenticate and access an **EC2 instance**.

It is mainly used for:

- **SSH access** to Linux EC2 instances.
- **RDP-related access** to Windows instances, including retrieving the initial Windows administrator password.
- Secure, passwordless authentication mechanisms.

A key pair consists of:

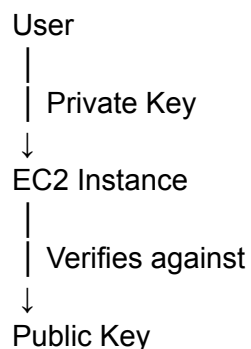
Public Key

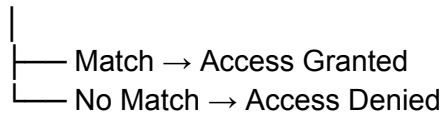
- Stored on/in association with the EC2 instance.
- Can be shared.
- Used to verify possession of the corresponding private key.
- AWS does **not** need the private key.

Private Key

- Given to the user when the key pair is created.
- Must be kept **secret**.
- Used by the user during authentication.
- AWS does **not** retain a copy that can be downloaded later.
- Losing it can make normal key-based access to an instance unavailable.

Authentication Flow





The private key **never needs to be transmitted to AWS as a password**. SSH uses public-key cryptography to prove that the user possesses the corresponding private key.

Why Key Pairs Are Important

Key-based authentication provides:

- Stronger authentication than weak passwords.
- Protection against password-guessing attacks.
- Secure administrative access.
- Support for automation and scripting.
- Standard SSH-based authentication for Linux systems.

Key Pair Best Practices

- **Never share the private key.**
- Store it securely with appropriate file permissions.
- Never commit it to GitHub or source control.
- Use separate keys where appropriate for different environments/users.
- Rotate or replace keys according to organizational security policies.
- Maintain secure backup/recovery procedures.
- Prefer centralized identity/access-management solutions where appropriate for larger environments.

Remember: The public key can be shared; the private key must be protected.

2. Introduction to Subnets

A **Subnet (Subnet = Subnetwork)** is a smaller network created inside a **VPC**.

The basic hierarchy is:

AWS Region



VPC



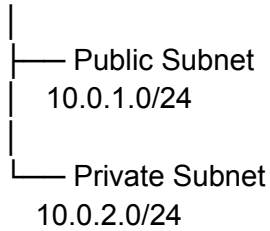
Subnet



AWS Resources

For example:

VPC: 10.0.0.0/16



Why Subnets Are Used

Subnets allow you to:

- Divide a VPC into smaller networks.
 - Organize resources.
 - Control network routing.
 - Separate public-facing and internal resources.
 - Improve network security and segmentation.
 - Design highly available architectures across Availability Zones.
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Public and Private Subnets

Public Subnet

A subnet is considered **public** when its routing configuration provides a path to an **Internet Gateway**.

Commonly used for:

- Internet-facing load balancers.
- Public-facing application components.
- Other resources that intentionally require direct internet connectivity.

Internet



Internet Gateway



Public Subnet

Private Subnet

A private subnet does **not** have a direct route to an Internet Gateway.

Commonly used for:

- Backend/application servers.
- Databases.
- Internal services.

Public Layer



Private Layer



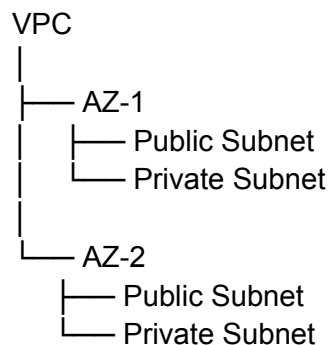
Database

A private subnet can still have **outbound internet connectivity through a NAT Gateway**, if required.

Important AWS Subnet Rule

One subnet belongs to exactly one Availability Zone.

However, a VPC can contain subnets across multiple Availability Zones:



This allows applications to be distributed across AZs for **high availability and fault tolerance**.

Key Takeaway

VPC = larger virtual network

Subnet = smaller network inside the VPC

Public subnet = has a route toward an Internet Gateway

Private subnet = no direct route toward an Internet Gateway
One subnet = one Availability Zone